



Charting the Future
of Global Health

Welcome to the IGHS AI Lab!

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August 18, 2026

An AI Lab (No Lab Coat Required)

- Solve real problems
- Discover something new
- Explore out loud
- Lead with curiosity
- Keep it fun
- <https://tinyurl.com/ighs-ai-lab>



Today's Agenda

- Presentation/demo: “Building a LitSearch Assistant GPT”
 - Eileen Chen, MLIS
 - UCSF Library
- Q&A

Upcoming

Date	Topic	Discussant(s)
9/1/2026	TBD	TBD
9/15/2026	Considering AI-enabled research and funding platforms	Gabe Murphy & Vincent Chen (OCR) and IGHS
9/29/2026	ChatGPT and MS Office plugins	Aaron McCoy, IGHS
10/13/2026	Using AI for QUAL research, opportunities & potential pitfalls	James Goh, Evidano



Upcoming AI Office Hours at IGHS

- Bi-weekly Tuesdays, 12-1pmPT
- Next is Tuesday, September 1
- Let me know if you want to present or lead a discussion topic
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LitSearch Assistant GPT

August 18, 2026

Eileen Chen, MLIS

Education and Research Librarian



Background

- Developed the Custom GPT **LitSearch Partner** with Yea-Hung Chen, Josephine Tan, and Elisa Jang for ChatGPT Hackathon
 - **Thought experiment:** how much of an Education and Research librarian's job can ChatGPT do?
- **New goals (simpler, more tailored):**
 - Help users learn about and develop database search strategies
 - Not just spew output, but involve users in search development
 - Publish to rest of UCSF as new available tool

Why search databases?

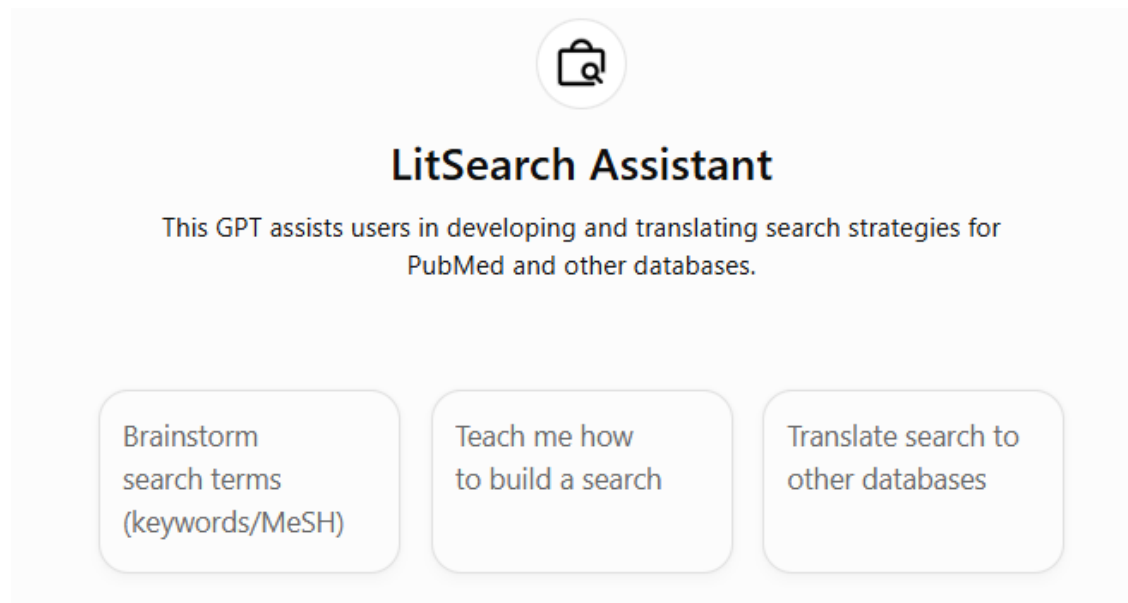
- **Controlled, reproducible** searches
- Expected for **evidence synthesis** projects (e.g. systematic reviews)
- Most AI tools (e.g. ChatGPT) retrieve a **curated subset of results** rather than run a comprehensive search
- AI tools may face **paywalls**

Functions

1. **Brainstorm** keywords/MeSH for a PubMed search
2. **Instruct** users on how to build a search strategy
3. **Translate** a search among 4 databases: PubMed, Embase, Web of Science, and CINAHL

Basic configuration

“This GPT was developed by a UCSF librarian and is intended for assisting users in developing and translating search strategies for PubMed and other databases. **[list of functions]** If at any point a user asks a question beyond what is covered in the following sections, refer them to the UCSF Library Help Center and/or suggest booking a consultation: <https://libraryhelp.ucsf.edu/>”



Brainstorm search terms

What are the impacts of extreme heat on peripartum maternal health in lower- and middle-income countries?

I'd break your question into **three main searchable concepts**:

Concept	What it represents
Extreme heat	Exposure to high ambient temperatures, heat waves, heat stress, or unusually hot weather
Peripartum maternal health	Maternal health during the period surrounding childbirth, potentially including late pregnancy, labor/delivery, and postpartum outcomes
Lower- and middle-income countries	Geographic/economic setting, including LMICs generally or specific countries/regions

Teaching search development

Teach me how to build a search

Let's start with the first building block of a PubMed search: **Boolean operators**. We'll go one step at a time.

Step 1: Boolean operators (AND, OR, NOT)

These operators tell PubMed how different search terms relate to each other.

Operator	Purpose	Example	Effect
AND	Finds records containing both concepts	diabetes AND exercise	Narrows results
OR	Finds records containing either term	adolescent OR teenager	Broadens results
NOT	Excludes records containing a term	asthma NOT pediatric	Narrows results by excluding

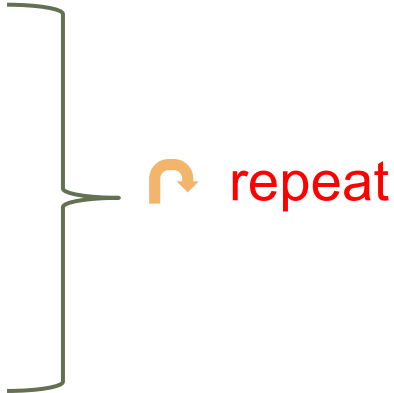
Search translation

Web of Science — complete translated strategy

```
TS=(  
  "extreme heat"  
  OR "hot temperature"  
  OR "hot temperatures"  
  OR "heat stress disorder*"  
  OR "heat stroke"  
  OR heatstroke  
  OR "heat wave"  
  OR "heat waves"  
  OR heatwave  
  OR heatwaves  
  OR "heat exposure"  
  OR "heat stress"  
  OR "heat illness"  
  OR "high temperature"  
  OR "high temperatures"  
  OR "ambient temperature"  
  OR "ambient temperatures"  
  OR "hot weather"  
  OR "thermal stress"  
)
```



Process

1. Outline objectives
 2. Break down objectives into defined steps
 3. Test (normal use)
 4. Revise
 5. Test (abnormal use)
 6. Revise
- 
- A diagram consisting of a large right-facing curly bracket that groups steps 3 through 6 of the list. To the right of the middle of the bracket is an orange circular arrow icon pointing clockwise, followed by the word "repeat" in red text.

Challenges & Solutions

Problem: GPT making mistakes it was explicitly told not to, e.g.

- Repeating something it already addressed
 - **Mitigation:** establish “**next**” as the command to move on
- Mistakes with translating search syntax
 - **Mitigation:** provide examples and explanations; add prominent disclaimer

Problem: hallucinations

- **Mitigation:** insist that it provides references; add prominent disclaimer

Problem: cannot modify temperature/tone

- **Mitigation:** define the tone (factual, concise) in free text at the start

Takeaways

1. **Operationalize** everything: definitions and steps are key
2. Ask it to **explain** repeated mistakes
3. Ask it to provide **links/references**
4. If sharing with others, add **disclaimers**
5. **Iterative process**: test, revise, and repeat

Next steps

- More testing of challenging topics and extended uses, e.g.
 - Option to “skip” when user is already familiar with search steps
 - Testing more “out of scope” questions

Questions?



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IGHS Video Tutorial for Creating a CustomGPT



<http://tiny.ucsf.edu/YPuX5b>

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Further IGHS Support: Chat Champions

- Andrea DeLuca
- CC Lipp
- Melissa Paschuck
- Josh Richardson
- Kelly Taylor
- Paul Wesson

On Teams at: [IGHS Chat Champions](http://tiny.ucsf.edu/vZmhYL)



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